EE 156: FUNDAMENTALS OF ELECTRICAL CONTROLS AND PLC SYSTEMS

Chapter 4: Schematic and Wiring Diagrams & Motor Starting

4.1 Introduction

Purpose of Diagrams

- Schematic and wiring diagrams are the **writing language of control circuits**
- Used to **connect** and **troubleshoot** control circuits
- Essential for understanding circuit operation

4.2 Schematic Diagrams

Definition

- Shows components in their **electrical sequence**
- **Without regard to physical location**
- Used to **connect** or **troubleshoot** a control circuit

Key Characteristics

Feature	Description
Electrical sequence	Shows how components connect electrically
Physical location	NOT shown
Physical size	NOT shown
Appearance	NOT shown

Purpose

- Helps technicians **trace circuit functions**
- Shows **logical sequence** of operation
- Essential for **troubleshooting**

Example: Start-Stop Push Button Circuit

```
      L1                    L2
       |                     |
       |                     |
+------+---------------------+------+
|      |                     |      |
|   +--+--+                  |      |
|   |     |                  |      |
|   | 1   |    +-+           |      |
|   |     |    | |           |      |
|   +--+--+    | |   +-----+ |      |
|      |       | |   |     | |      |
|      +-------+-+   |  M  | |      |
|              | |   |     | |      |
|   +--+--+    | |   +-----+ |      |
|   |     |    | |           |      |
|   | 2   |    +-+           |      |
|   |     |                  |      |
```

```
|    +--+--+                         |        |
|         |                           |        |
+------+-------------------+-------+
         |                           |
         v                           v
```

4.3 Wiring Diagrams

Definition

- Shows **physical layout** of equipment
- Shows **wire connections** between devices
- Shows **relative layout** of components

Key Characteristics

Feature	Description
Physical location	Shows relative layout
Wire connections	Shows conductor terminations
Physical relation	Shows how devices are arranged

Use

- Motor control installations
- Equipment installation
- Field service and maintenance

Wiring Diagram vs Schematic

Feature	Schematic Diagram	Wiring Diagram
Shows	Electrical sequence	Physical layout
Location	Not shown	Shown
Size/appearance	Not shown	Shown
Use	Troubleshooting	Installation
Understanding	Circuit function	Physical connections

4.4 Basic Schematic Rules

Rule 1: Reading Direction

Read a schematic as you would a book:

- From **top to bottom**
- From **left to right**

Rule 2: De-Energized State

Schematics show components in their:

- **De-energized** state
- **OFF** state
- Normal (unpowered) position

Rule 3: Contact Labels

- Any contact with the **same label or number** as a coil
- Is **controlled by that coil**

Rule 4: Contact Operation

When a coil **energizes**:

- **NO contacts** $\rightarrow$ **CLOSE**
- **NC contacts** $\rightarrow$ **OPEN**

When a coil **de-energizes**:

- Contacts **return to normal state**

4.5 Ladder Logic Diagrams

Definition

- Used to show **relationships** between circuit components
- **Not** physical location
- Provides **fast, easy understanding** of circuit connections

Components

- Uses **graphic symbols** (industry standards)
- Uses **abbreviations**
- Uses **device designations**

Ladder Diagram Structure

```
[LIVE Rail]              [NEUTRAL Rail]
     |                        |
     |                        |
+----+------------------------+----+
|    |         RUNG 1         |    |
|   [Device]---[Device]---[Load]   |
|    |                        |    |
|    |         RUNG 2         |    |
|   [Device]---[Device]---[Load]   |
|    |                        |    |
|    |         RUNG 3         |    |
|   [Device]---[Device]---[Load]   |
```

```
|    |                         |    |
+----+-------------------------+----+
     |                         |
     v                         v
```

4.6 Ladder Diagram Rules

Rule 1: Reading Direction

- Read from **left to right**
- Then from **top to bottom**
- Understand the **sequence of operation**

Rule 2: Load Placement

- **Loads** shall be connected in **parallel on the rungs**
- Last component connected to right side of rail

Rule 3: Load Position

- Load is the **last component** connected to the right rail
- Exception: **Protective contacts** (overload) can be after load

Rule 4: Control Device Placement

Control devices are connected:

- Between **left rail and the load**
- Or between **other control devices and the load**

Rule 5: Short Circuit Prevention

Control devices are **NEVER** connected:

- From **left rail to right rail**
- This creates a **short circuit**

Rule 6: Series and Parallel Connections

Function	Connection Type
STOP functions	Connected in SERIES
START functions	Connected in PARALLEL

4.7 Power Circuit

Three-Phase Motor Power Circuit Components

Component	Description
3PST Disconnect	Three-pole single-throw disconnect switch
Overcurrent Protection	Fuses or circuit breakers
Motor Starter Contacts	Horsepower-rated contacts
Overload Relay Heaters	Thermal protection elements

Typical Power Circuit

```
      L1     L2     L3
       |      |      |
       |      |      |
    +--+--+ +--+--+ +--+--+
    |     | |     | |     |    Disconnect Switch
    +--+--+ +--+--+ +--+--+
       |      |      |
    +--+--+ +--+--+ +--+--+
    |     | |     | |     |    Fuses
    +--+--+ +--+--+ +--+--+
       |      |      |
    +--+--+ +--+--+ +--+--+
    |     | |     | |     |    Contactor Contacts
    +--+--+ +--+--+ +--+--+
       |      |      |
    +--+--+ +--+--+ +--+--+
    |     | |     | |     |    Overload Heaters
    +--+--+ +--+--+ +--+--+
       |      |      |
       v      v      v
      M1     M2     M3     (Motor)
```

4.8 Interpretation of Schematic Diagrams

Example 1: Pressure Switch Control Circuit

Circuit Description:

- Pressure switch connected to **control relay (CR)**
- If pressure becomes **too great:**
  - **CR** opens **NC contact** to motor starter → Motor **STOPS**
  - **CR** closes **NO contact** → Pilot light turns **ON**
  - **High-pressure condition** indicated

Components:

- Mercury pressure switch
- Control relay (CR)
- Motor starter coil
- Pilot light

Operation Sequence:

```
Normal Pressure -> Pressure Switch Open -> Motor Running
                       |
High Pressure -> Pressure Switch Closes -> CR Energizes
                       |
    +------------------+------------------+
    |                                     |
CR NC Contact Opens                  CR NO Contact Closes
    |                                     |
Motor Stops                          Warning Light ON
```

Example 2: Tank Level Control with Float Switches

Circuit Components:

- **FS1** - Lowest level detection (turns both pumps OFF)
- **FS2** - Starts Pump #1 when liquid reaches height
- **FS3** - Starts Pump #2 if level continues rising
- **FS4** - Warning light and buzzer (overflow warning)

Operation Sequence:

```
Level Rising:
    FS2 activates -> Pump #1 Starts
         | (if level continues rising)
    FS3 activates -> Pump #2 Starts
         | (if level continues rising)
    FS4 activates -> Warning Light + Buzzer ON

Level Falling:
    FS1 activates -> Both Pumps Stop
```

Reset Function:

- Reset button turns off buzzer
- Warning light remains ON until water level drops below FS4

Float Switch Arrangement:

```
    [Overflow Level]
         |
    FS4 -+--- Warning Light & Buzzer
         |
    FS3 -+--- Pump #2 Start
         |
    FS2 -+--- Pump #1 Start
         |
    FS1 -+--- Both Pumps Stop
         |
    [Bottom Level]
```

Example 3: Air Conditioning Control Circuit

Components:

- **Thermostat** - Temperature control
- **CR** - Control relay
- **CFM** - Condenser fan motor relay
- **EFM** - Evaporator fan motor relay
- **Comp.** - Compressor relay
- **Air flow switches** - Safety interlocks

Operation Sequence:

1. **Thermostat Calls for Cooling:**

```
Thermostat Closes -> CR Energizes
                        |
        +---------------+---------------+
        |               |               |
    CFM Energizes   EFM Energizes   Compressor
        |               |           (Inhibited)
    Condenser Fan   Evaporator Fan
    Starts          Starts
```

2. **Air Flow Established:**

```
Condenser Fan Starts -> Air Flow Switch #1 Closes
Evaporator Fan Starts -> Air Flow Switch #2 Closes
                        |
                    Both Switches Closed
                        |
                    Compressor Relay Energizes
                        |
                    Compressor Starts
```

Safety Feature: Compressor cannot start until **BOTH** air flow switches confirm fan operation.

Example 4: Elevator/Platform Control Circuit

Circuit Components:

- **LS1** - Limit switch (top floor) - NC
- **LS2** - Limit switch (bottom floor) - NC
- **M** - Motor contactor
- **CR** - Control relay
- **UP** - Up push button
- **DOWN** - Down push button

Initial State:

- Platform at bottom floor
- LS2 is **OPEN** (actuated)
- LS1 is **CLOSED**

Up Operation:

```
Press UP Button -> M Energizes
                      |
              Motor Starts (Raising)
                      |
         M NC Contact Opens (CR Interlocked)
                      |
         Platform Rises -> LS2 Closes
                      |
         Continues until LS1 Actuates
                      |
              LS1 Opens -> M De-energizes
                      |
            Platform Stops at Top
```

Down Operation:

```
Press DOWN Button -> CR Energizes
                        |
    +-------------------+-------------------+
    |                                       |
CR NC Contact Opens               CR NO Contact Closes
(M Interlock)                    (Solenoid Energizes)
    |                                       |
M Cannot Energize                 Platform Moves Down
                                        |
                              LS1 Re-closes (as platform leaves top)
                                        |
                              Continues until LS2 Opens
                                        |
                            Platform Stops at Bottom
```

Interlocking:

- M and CR are **interlocked**
- Only **one can be energized at a time**
- Prevents simultaneous UP and DOWN commands

Example 5: Class Exercise - Describe Circuit Operation

Given:

- Line H = Live line (positive rail)
- Control circuit with multiple components

Question: Describe the operation of the control diagram.

Approach:

1. Identify all components
2. Trace circuits from left to right
3. Determine sequence of operation
4. Identify interlocking and safety features
5. Explain control logic

4.9 Design Considerations for Ladder Diagrams

Step-by-Step Design Process

Step	Action
1	Define the process to be controlled
2	Draw a sketch of the operation process
3	Ensure all components are present in drawing
4	Determine sequence of operations
5	List sequence steps in detail (sentences or table)
6	Write relay logic diagram from sequence

4.10 Design Examples

Circuit #1: Pressure Detection & Control Circuit

Requirement: Design a circuit that:

1. Sounds a horn and turns on red warning light when pressure becomes extreme
2. Button can turn horn OFF but keep warning light ON
3. Warning light stays ON until pressure drops to safe level

Solution (schematic layout):

```
L1                                              L2
 |                                               |
 |    +-----+    +-------+    +-----------------+|
 |    |     |    |       |    |                 ||
 +----+ PS  +----+  CR   +----+   Alarm Horn    ||
 |    |     |    |       |    |                 ||
 |    +--+--+    +---+---+    +-----------------+|
 |       |           |                            |
 |       |    +------+------+    +---------------+|
 |       |    |             |    |               ||
 |       +----+ CR NO       +----+ Red Warning   ||
 |            |   (Seal)    |    | Light         ||
 |            |             |    |               ||
 |            +-------------+    +---------------+|
 |                                               |
 |            +-------------+    +---------------+|
 |            |             |    |               ||
 +------------+ Reset PB    +----+ Horn Circuit  ||
              |   (NC)      |    |               ||
              |             |    |               ||
              +-------------+    +---------------+|
                                                  |
L1                                              L2
```

Operation:

1. **Normal Condition:**
   - Pressure switch (PS) is OPEN
   - CR is de-energized
   - No alarm
2. **Pressure Becomes Extreme:**
   - PS closes
   - CR energizes
   - CR NO contact closes (seal circuit)
   - Horn sounds
   - Red warning light turns ON
3. **Pressing Horn OFF Button:**
   - Breaks horn circuit only
   - Red warning light remains ON (seal circuit maintains)
4. **Pressure Drops to Safe Level:**
   - PS opens
   - CR de-energizes
   - Red warning light turns OFF
   - Circuit resets for next event

Circuit #2: Lamp Control Circuit

Requirement: Design a circuit with:

1. STOP/START push buttons
2. Red and green bulbs
3. Red bulb ON when circuit energized
4. Green bulb OFF until START pressed
5. When green light comes ON, red goes OFF
6. Only STOP button can turn green OFF

Solution (schematic layout):

```
L1                                              L2
 |                                               |
 |    +-----+    +--------+   +-----------------+|
 |    |     |    |        |   |                 ||
 +----+START+----+   CR   +---+   Green Light   ||
 |    |     |    |        |   |                 ||
 |    +--+--+    +---+----+   +-----------------+|
 |       |           |                            |
 |       |    +------+------+                     |
 |       |    |             |                     |
 |       +----+ CR NO       |                     |
 |            |   (Seal)    |    +---------------+|
 |            |             |    |               ||
 |            +-------------+    |   Red Light   ||
 |                                |               ||
 |            +-------------+    |               ||
 |            |             |    |               ||
 +------------+ STOP        +----+               ||
              |   (NC)      |    |               ||
              |             |    |               ||
              +-------------+    +---------------+|
                                                  |
L1                                              L2
```

Operation:

1. **Initial State (Energized):**
   • Red light is ON (connected directly across supply)
   • Green light is OFF
2. **START Button Pressed:**
   • CR energizes
   • CR NO contact closes (seals)
   • Green light turns ON
   • Red light turns OFF
3. **START Button Released:**
   • CR remains energized (sealed through its own contact)
   • Green light stays ON
   • Red light stays OFF
4. **STOP Button Pressed:**
   • CR de-energizes
   • CR NO contact opens
   • Green light turns OFF
   • Red light turns ON
   • Circuit returns to initial state

4.11 Key Terminology

Term	Definition
Schematic	Electrical sequence diagram
Wiring diagram	Physical layout and connections
Ladder diagram	Control circuit logic representation
Rails	Vertical power lines (L1/L2)
Rungs	Horizontal circuit branches
Load	Device that consumes power
Control device	Switch, relay, timer that controls circuit
Seal circuit	Maintains circuit after momentary input
Interlock	Prevents conflicting operations
De-energized	Powered-off state

Review Questions

1. What is the difference between a schematic diagram and a wiring diagram?
2. Why are schematics drawn with components in their de-energized state?
3. What happens to NO and NC contacts when a coil energizes?
4. Explain the structure of a ladder diagram (rails and rungs).
5. Where should loads be connected in a ladder diagram?
6. Why should control devices never be connected from left rail to right rail?
7. How are STOP and START functions typically connected in ladder diagrams?
8. What is a seal circuit and why is it used?
9. Explain the purpose of interlocking in control circuits.
10. Describe the sequence for designing a ladder logic diagram.

END OF CHAPTER 4 NOTES